



Decades running trades operations — crews, assets, P&L, growth — and for the past two years the one-person AI department inside a live services company: agents wired into a real CRM, a company knowledge engine, voice agents, vision pipelines, and the production plumbing between them. Nothing here is a demo; most is in daily use. Client and company names withheld by agreement — the architecture is the résumé.

◆ SYSTEMS SHIPPED · PRODUCTION

Company knowledge engine (“DMF Brain”) MCP · pgvector · RRF · human-gated

Captures from every work surface through one MCP agent protocol; staged, human-gated ingestion; hybrid RRF-fused retrieval; append-only decision ledger; watcher agents flag cross-project reuse; served to models via a 9-tool MCP server.

Agent platform Langfuse · guardrails · state machines

Agents as governed infrastructure: one spec shape (tools, outcome, guardrails, tests), a shared harness with loop guards/budgets/retries, run-by-run observability, and a human-approval spine on anything touching money or customers.

CRM intelligence layer Salesforce · Apex · LWC · Flow · CPQ

Stage-triggered briefs from parallel queries across 15+ objects; a conversational agent with live SOQL-backed tools over real orders; plus CPQ, payments, and e-signature. Used by a real sales team daily.

Vision & document pipelines vision + structured JSON · serverless

Receipt photos to categorized ledger rows; serverless PDF estimate extraction; automated quality flags on field photos. Every pipeline validates, retries, and fails loudly.

Voice agent pipeline research → complete build spec

Researches a business, derives what its phone line must do, and emits the complete build — configs, flows, tools, knowledge base — delivered as build-spec microsites.

◆ PRODUCT LINE · THE DMF COMPANY

- ◆ **ServiceRelay** — voice + web booking/intake; answers, books, follows up; white-label.
- ◆ **Prospecting & lead engine** — farm → enrich → score → reach; human-approval spine.
- ◆ **Image generation array** — brief → on-brand image sets; style-locked, multi-model.
- ◆ **Video generation pipeline** — script → scenes → ad-ready cuts; closes the loop on ad performance.
- ◆ **Competitive intelligence** — SEO/SERP feeds → warehouse + live dashboards, scoped to one market.
- ◆ **Marketing automation suite** — full-funnel, run by role-agents under one brand memory.

◆ METHOD — ONE OPERATOR, TEAM-SCALE OUTPUT

- ◆ A 40-skill library encodes conventions, schemas, brand systems, error playbooks, and QA gates.
- ◆ Tools as hands: MCP connectors to database, deploy target, automation, and comms.
- ◆ Proof over claims: headless-browser harnesses and numeric render gates (incl. a 24-check 3D QA pipeline).
- ◆ Structured handoffs + append-only decision logs keep long builds honest.
- ◆ Build loops run their own tests; watcher agents transfer advances between projects.
- ◆ Exact-fit tooling over heavy platforms — full-stack by necessity, simple by conviction.

◆ STACK

Models & agents Claude API (tools, structured output, vision), MCP servers/clients, Langfuse, agent SDKs.	Data & records Supabase/Postgres (pgvector, RLS, Edge Functions), Salesforce (Apex, LWC, Flow, CPQ).
Automation & comms n8n, Twilio (voice/SMS), ElevenLabs, Stripe.	Web & 3D three.js / R3F, React · Astro · Vite, Netlify, headless Chromium QA rigs.

◆ BACKGROUND — THE EDGE

- ◆ Journeyman first, engineer by necessity — ran the business before writing the code that runs it.
- ◆ Managed crews, client relationships, and vendor networks across B2B and B2C; KPI-driven, gamified performance systems.
- ◆ Then built the full-stack, custom, automated software those operations actually needed. Day one: full toolkit, zero ramp.